Task Order

Project Name: A-SW Hub Platform Development **Contract Period**: September 1, 2025 ~ October 25, 2025 (55 days) **Contracting Agency**: Korea Institute of Industrial Technology (KITECH) **Performing Organization**: VIA Corporation **Document Number**: TO-2025-ASW-001 **Date**: September 1, 2025 **Version**: 1.0

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1. Background and Objectives

1.1 Background

Developing autonomous agricultural machinery software (A-SW) is a complex task requiring convergence of various specialized technologies, necessitating an efficient collaboration system among five participating institutions (KITECH, TYMICT, JBNU, VIA, OntarioTech).

Traditional collaboration methods using email and file sharing have the following limitations:

- Delayed development cycles (3-6 months) due to limited real-time collaboration
- Version conflicts and quality issues during code integration
- Inefficiencies from manual build/deployment processes
- Difficulty tracking project progress

1.2 Objectives

This task aims to build an **integrated collaboration platform (A-SW Hub)** for autonomous agricultural machinery software development, establishing a real-time collaboration system among participating institutions and improving development efficiency by more than 75%.

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2. Scope of Work

2.1 Core Features

2.1.1 Authentication & Authorization

- Email/Password authentication + GitHub OAuth 2.0 integration

- Role-Based Access Control (RBAC): Admin, Developer, Viewer
- JWT-based session management with automatic renewal

2.1.2 Project Management

- Integrated management of agricultural machinery A-SW projects
- Git repository integration (GitHub/GitLab)
- Project settings: General, Repository, Build, Deployment, Notifications
- Collaborator management and permission settings

2.1.3 CI/CD Pipeline

- Automatic build triggers (Git Push → Build → Test → Deploy)
- Real-time build log streaming
- 9-stage build process visualization
- Automatic notifications and rollback on build failures

2.1.4 Deployment Management

- Multi-stage deployment environments: Development, Staging, Production
- Deployment approval workflows
- Deployment history tracking

2.1.5 Quality Management

- Code quality metrics: Coverage, Complexity, Duplication
- Multi-level approval workflows (2-tier approval)
- Automatic quality gate verification

2.1.6 Dashboard & Monitoring

- Real-time KPI dashboard
- Build success rate, deployment frequency, average lead time tracking
- Activity feed and notifications
- Project distribution visualization by language

2.1.7 Timeline Management

- 5-year plan Gantt chart visualization

- Progress tracking of 136 deliverables
- Hierarchical structure by institution/year
- Milestone and dependency management

2.2 Technical Requirements

- **Frontend**: SvelteKit 2.x, TypeScript 5.x (Strict Mode)
- **State Management**: Svelte Stores
- **Validation**: Zod (runtime type validation)
- **Visualization**: Chart.js
- **Design**: Extreme Black & White Industrial Design
- **Testing**: Vitest (Unit), Playwright (E2E)
- **Build**: Zero TypeScript Errors, Zero Warnings

2.3 Non-Functional Requirements

- **Performance**: Lighthouse scores Performance 95+, Accessibility 100
- **Security**: HTTPS Only, OAuth 2.0, JWT, RBAC, CORS, CSP
- **Accessibility**: WCAG 2.1 AA compliant
- **Browsers**: Chrome 90+, Safari 14+, Firefox 88+
- **Responsive**: Desktop (1024px+), Tablet (768px+), Mobile (< 768px)

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3. Deliverables

3.1 Software Deliverables

No.	Deliverable	Format	Notes
1	A-SW Hub Platform Source Code	Git Repository	GitHub/GitLab
2	Production Build Package	ZIP	Static files
3	Environment Configuration (.env.example)	ENV	With deployment guide

3.2 Documentation Deliverables

No.	Document	Format	Notes
4	System Architecture Design	Markdown/PDF	Korean/English
5	API Documentation	Markdown	RESTful endpoints
6	User Manual	Markdown/PDF	Korean/English, with screenshots
7	Administrator Guide	Markdown/PDF	Installation, configuration, operations
8	Design System Documentation	Markdown	B&W Industrial Design
9	Final Report	PDF	Korean/English

3.3 Training Materials

No.	Material	Format	Notes
10	Onboarding Training Program	Presentation	< 10 minutes
11	Hands-on Practice Guide	Markdown	Step-by-step

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4. Schedule

4.1 Overall Schedule (55 days)

Period	Phase	Key Activities	Deliverables
09.01~09.15 (15 days)	Requirements Analysis & Design	- System architecture design - UX/UI design guide - Development environment setup	Design documents, Design guide
09.16~09.30 (15 days)	Core Feature Development	- Auth/Authorization system - Project management module - CI/CD pipeline (basic)	Feature implementation complete

Period	Phase	Key Activities	Deliverables
10.01~10.15 (15 days)	Advanced Feature Development	- Dashboard & metrics - Quality management system - Timeline Gantt chart	Advanced features complete
10.16~10.25 (10 days)	Testing & Documentation	- Integration testing - User manual - Final report	All deliverables

4.2 Major Milestones

Date	Milestone	Review Items
09.15	Design Complete	Architecture design, Design guide review
09.30	Core Features Complete	Auth, Project management, Basic CI/CD demo
10.15	All Features Complete	Integrated demo of all features
10.25	Final Delivery	Acceptance inspection of all deliverables

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5. Roles and Responsibilities

5.1 Contracting Agency: Korea Institute of Industrial Technology (KITECH)

- Overall task management and coordination
- Requirements review and approval
- Interim/final inspections
- Coordination among participating institutions

5.2 Performing Organization: VIA Corporation

- Complete design and development of A-SW Hub platform
- CI/CD pipeline construction
- Black & White Industrial Design System implementation
- TypeScript-based enterprise architecture implementation

- Complete documentation (Korean/English)
- Onboarding training program development and delivery

5.3 Collaborating Organizations

- **TYMICT, JBNU, OntarioTech**: Platform testing and feedback

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6. Acceptance Criteria

6.1 Quantitative Criteria

Item	Target	Measurement Method
Build Success Rate	100%	CI/CD build log verification
TypeScript Type Coverage	100%	Strict Mode compilation success
Development Cycle Reduction	75%	Validate 3-6 months → 2-4 weeks
Lighthouse Performance	95+	Chrome DevTools measurement
Lighthouse Accessibility	100	Chrome DevTools measurement
Bundle Size	< 200KB (gzipped)	Build analysis
Unit Test Coverage	85%+	Vitest coverage report

6.2 Qualitative Criteria

Item	Inspection Method
Feature Completeness	Verify 100% implementation vs requirements spec
Usability	Real-world testing based on user scenarios
Code Quality	Code review, SonarQube analysis
Documentation Completeness	Document review and practice feasibility check
Design Consistency	Extreme B&W Industrial Design compliance

6.3 Acceptance Conditions

All of the following conditions must be met for acceptance:

1. ✅ All functional requirements implemented (per requirements specification)
2. ✅ All quantitative criteria achieved
3. ✅ All deliverables submitted (software, documentation, training materials)
4. ✅ Integration testing passed (100% of E2E critical flows)
5. ✅ Contracting agency approval

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7. Reference Documents

Please refer to the following documents for task execution:

Document	Filename	Purpose
Requirements Specification	`requirements-spec.md`	Detailed functional/non-functional requirements
User Scenarios	`user-scenarios.md`	Actual usage flows and test scenarios
Architecture Guide	`architecture-guide.md`	Technical architecture and design principles

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8. Special Notes

8.1 Design Principles

- **Extreme Black & White Industrial Design**: Only #000000 and #FFFFFF, border-radius: 0
- **Configuration-Driven Architecture**: Environment variable-based control
- **Domain-Driven Design**: Independent module design by domain
- **Zero-Touch Deployment**: Fully automated from code push to deployment

8.2 Out of Scope

The following are **excluded** from this task:

- Actual agricultural machinery hardware control interfaces
- AI/ML model training and inference engines
- Real-time sensor data collection systems
- Native mobile apps (replaced with responsive web)
- Backend API server construction (uses mock data)

8.3 Intellectual Property Rights

- All intellectual property rights for deliverables belong to the **contracting agency (KITECH)**.
- External libraries with open-source licenses follow their respective licenses.

8.4 Change Management

- Scope changes require written agreement between contracting and performing organizations
- Schedule delays must be reported immediately with adjustment discussions

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9. Contract Terms (Reference)

> This section is separately defined in the actual contract and is included here for reference only.

9.1 Task Compensation

- Total task amount: [Separate contract]
- Payment terms: [Separate contract]

9.2 Progress Reporting

- Weekly progress report: Every Friday (email)
- Monthly review meeting: End of each month (in-person/video)

9.3 Inspection and Delivery

- Interim inspection: October 15, 2025 (when all features development complete)
- Final inspection: October 25, 2025 (when all deliverables submitted)

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10. Approval

Role	Organization	Name	Signature	Date
Contracting Agency Lead	KITECH			2025.09.01
Performing Organization Lead	VIA			2025.09.01

Document History

Version	Date	Author	Changes
1.0	2025.09.01	VIA	Initial creation

Appendices

- Appendix A: Requirements Specification (`requirements-spec.md`)
- Appendix B: User Scenarios (`user-scenarios.md`)
- Appendix C: Architecture Guide (`architecture-guide.md`)